

Bayesian data fusion developments for causal inference

Augmenting clinical trials with real-world observational evidence

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Paper & slides

- **Motivation:** the data fusion dilemma (RCTs vs. RWD)
- **Foundational baseline:** power priors and normalization challenges
- **Contrast-level hierarchical borrowing:**
 - Robust meta-analytic-predictive (MAP) priors ($H \geq 3$)
 - Pairwise shrinkage for single historical studies ($H = 1$)
 - Variance inflation as power likelihood discounting
- **Individual-level data fusion (Lin et al., 2026):**
 - Joint power likelihoods and frugal parameterization
 - Out-of-sample predictive optimization (ELPD_{cv})
 - Asymptotics, sandwich covariance, and privacy
- **Methodological synthesis and open frontiers**

The clinical trade-off:

- *Randomised controlled trials (RCTs)*:
 - Gold standard for internal validity through design ($Y(t) \perp\!\!\!\perp T \mid W, S = e$).
 - Small sample sizes, high costs, restrictive criteria: limited statistical power and poor generalisability.
- *Observational real-world data (RWD)*:
 - Large patient cohorts, clinical registries, routine practice records.
 - Vulnerable to unmeasured confounding and selection bias.

Objective of data fusion (Bareinboim & Pearl, 2016)

Borrow statistical power from large observational cohorts while preserving the internal validity of the randomized trial.

The Bayesian paradigm offers a framework to introduce auxiliary evidence into prior distributions and adaptively discount it under data conflict.

Let $D_0 = (n_0, y_0, X_0)$ denote historical data with likelihood $L(\theta \mid D_0)$, and $\pi_0(\theta \mid c_0)$ an initial prior indexed by c_0 .

Power prior and posterior formulation

For a discounting hyperparameter $a_0 \in [0, 1]$:

$$\pi(\theta \mid D_0, a_0) \propto L(\theta \mid D_0)^{a_0} \pi_0(\theta \mid c_0), \quad \pi(\theta \mid D_e, D_0, a_0) \propto L(\theta \mid D_e) \pi(\theta \mid D_0, a_0)$$

- $a_0 = 1$ corresponds to unpenalized pooling; $a_0 = 0$ excludes external data.
- Disperses prior concentration around historical estimates to mitigate conflict.

Core limitations:

- *Fixed* a_0 : Purely subjective elicitation; no internal commensurability check.
- *Random* a_0 : Naive joint prior $\pi(\theta, a_0 \mid D_0) \propto L^{a_0} \pi_0 \pi(a_0)$ violates the likelihood principle (Duan et al., 2006).

Proper power prior (Duan et al., 2006; Neuenschwander et al., 2009)

Retaining $C(a_0) = [\int_{\Theta} L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) d\theta]^{-1}$ restores probability calculus:

$$\pi(\theta, a_0 | D_0) = C(a_0) L(\theta | D_0)^{a_0} \pi_0(\theta | c_0) \pi(a_0) \quad (a_0 \in A)$$

- *Prevents collapse*: Without $C(a_0)$, $\pi(a_0 | \text{data})$ collapses to 0 even for identical datasets.
- *Dynamic commensurability*: Concordance pushes $a_0 \rightarrow 1$; conflict forces $a_0 \rightarrow 0$.
- *Multi-study extension*: Assigns vector weights $a_0 = (a_{01}, \dots, a_{0H})^T$ for $H \geq 2$.

The single-source bottleneck ($H = 1$)

With only one historical study ($H = 1$), data cannot identify cross-study heterogeneity:

$\pi(a_0 | \text{data})$ remains sensitive to its prior.

Schmidli et al. (2014): Shift to hierarchical predictive borrowing for the control response parameter $\theta = g(\psi)$ across $H \geq 3$ historical studies.

Hierarchical exchangeability model

$$\theta_1, \dots, \theta_H, \theta_* \mid \mu, \tau \stackrel{\text{i.i.d.}}{\sim} \text{N}(\mu, \tau^2), \quad \pi(\mu, \tau) = \pi(\mu)\pi(\tau)$$

where $\tau \geq 0$ quantifies between-trial heterogeneity.

- *Conjugate mixture:* Predictive density $\pi(\psi_* \mid \text{data}) \approx \sum_{k=1}^K w_k \pi_c(\psi_* \mid \xi_k)$ via EM algorithm provides closed-form posterior updates during trial analysis.
- *Robustification:* Augment mixture with a vague component $\pi_V(\psi_*)$ weighted by $w_R \in [0.10, 0.20]$ (Cromwell's rule):

$$\hat{\pi}_{\text{HR}}(\psi_*) = (1 - w_R) \sum_{k=1}^K w_k \pi_c(\psi_* \mid \xi_k) + w_R \pi_V(\psi_*)$$

- *Adaptive discounting:* Under prior-data conflict ($m_k \ll m_V$), the Bayes factor ratio drives $\tilde{w}_R \rightarrow 1$, automatically discarding external controls.

- *Pairwise synthesis* ($H = 1$): Frequentist random-effects often yield $\hat{\tau}^2 = 0$, underestimating variance uncertainty.
- Symmetric exchangeability around an artificial mean μ is hard to justify.

Asymmetric reference model ($\theta_e = \mu$)

Anchors directly to the randomized trial target parameter:

$$\theta_e = \mu, \quad \theta_o \mid \mu, \tau \sim N(\mu, 2\tau^2), \quad k = \sqrt{2}\tau, \quad \tau \sim \text{HN}(s)$$

Conditional on τ , the shrinkage expectation for the trial parameter is

$$E\{\theta_e \mid \hat{\theta}_e, \hat{\theta}_o, \tau\} = \frac{\sigma_e^{-2}\hat{\theta}_e + (\sigma_o^2 + 2\tau^2)^{-1}\hat{\theta}_o}{\sigma_e^{-2} + (\sigma_o^2 + 2\tau^2)^{-1}}$$

- Integrates over $\pi(\tau \mid \hat{\theta}_e, \hat{\theta}_o)$, fully propagating heterogeneity uncertainty.
- As conflict increases ($\tau^2 \rightarrow \infty$), observational borrowing vanishes.

Analytical equivalence (Röver & Friede, 2020)

Equating observational shrinkage precision to power prior precision $a_0\sigma_o^{-2}$:

$$a_0\sigma_o^{-2} = \frac{1}{\sigma_o^2 + 2\tau^2} \iff a_0 = \frac{\sigma_o^2}{\sigma_o^2 + 2\tau^2} = \frac{1}{1 + 2\frac{\tau^2}{\sigma_o^2}}$$

Hierarchical variance inflation by $2\tau^2$ is algebraically equivalent to power likelihood discounting by $a_0 \in [0, 1]$.

Methodological caveats:

- ➊ *Asymptotic normality*: Requires Gaussian summary estimates; fragile for rare-event binary outcomes.
- ➋ *Dispersion vs. bias*: Symmetric offset $N(0, 2\tau^2)$ models random variation, not directional bias; observational data must be rigorously pre-adjusted.
- ➌ *Aggregate evidence*: Operates strictly on contrast summaries, lacking patient-level covariate granularity.

Lin et al. (2026): Combine experimental trial (X_e) and observational cohort X_o at the individual participant level $X_i = (Y_i, T_i, W_i)$.

Joint power likelihood

With domain indicator $S_i \in \{e, o\}$ and discounting rate $\eta \in [0, 1]$:

$$\pi(\phi \mid X_e, X_o, \eta) \propto f_e(X_e; \phi) \{f_o(X_o; \phi)\}^\eta \pi(\phi)$$

Structured parameter partitioning: $\phi = (\theta, \gamma, \psi) \in \mathbb{R}^p$

($p = p_\theta + p_\gamma + p_\psi$):

- $\theta \in \mathbb{R}^{p_\theta}$: Shared causal treatment effect contrast (parameter of interest).
- $\gamma \in \mathbb{R}^{p_\gamma}$: Trial-specific nuisance parameters.
- $\psi \in \mathbb{R}^{p_\psi}$: Observational nuisance parameters (confounding/selection).

Flexibility: Continuous discounting ($\eta = 0$: trial only; $\eta = 1$: full pooling) accommodates both single-arm (control) and two-arm observational data.

Frugal MSM factorisation (Evans & Didelez, 2024)

Partition covariates $W = (C, Z)$ into effect modifiers C and auxiliary Z :

$$f_s(y, t, w) = f_s(t, z | c) f_s(c) \cdot f_{Y(t)|C}^*(y | t, c; \theta) \cdot c_s^*(y, t, z | c; \zeta_s)$$

f^* models causal CATE $\tau(c)$; copula c_s^* absorbs domain nuisance ($\zeta_e = \gamma, \zeta_o = \psi$) without feedback onto θ .

Causal identification pillars:

- 1 *Randomization*: Ensures trial ignorability and positivity ($Y(t) \perp\!\!\!\perp T | W, S = e$).
- 2 *Representativeness*: $E\{Y(1) - Y(0) | C = c, S = e\} = \tau(c)$.
- 3 *Contrast transportability*: $E\{Y(1) - Y(0) | C = c, S = o\} = \tau(c)$.
- 4 *Model specification*: Correct causal MSM for counterfactuals $Y(t) | C$.

Adaptive selection: $\hat{\eta} = \arg \max_{\eta \in \mathcal{H}} \widehat{\text{ELPD}}_{\text{cv}}(\eta)$ evaluated on held-out RCT folds $D_e^{(k)}$. It preserves trial internal validity: unmeasured residual confounding forces $\hat{\eta} \rightarrow 0$.

Bernstein–von Mises asymptotic decoupling

As $n_e, n_o \rightarrow \infty$, the joint posterior converges to a Gaussian distribution:

$$\pi(\phi \mid X_e, X_o, \eta) \sim \text{N} \left(\hat{\phi}_\eta, (n_e V_e^{-1} + \eta n_o V_o^{-1})^{-1} \right)$$

with precision-weighted mean

$$\hat{\phi}_\eta = (n_e V_e^{-1} + \eta n_o V_o^{-1})^{-1} (n_e V_e^{-1} \hat{\phi}_e + \eta n_o V_o^{-1} \hat{\phi}_o), \text{ with sandwich } V_e, V_o.$$

- *MCMC-free optimization:* Closed-form Gaussian density allows rapid grid search.
- *Privacy preservation:* Participating sites share only summaries $(\hat{\phi}_s, V_s)$ without pooling raw individual patient data.

Methodology	Data level	Cohort setting	Commensurability	Confounding protection
Modified power prior (<i>Duan et al., 2006</i>)	Aggregate / IPD	Multi-study $H \geq 2$	Marginal likelihood Mode $a_0 \rightarrow 1$	Prior-level likelihood discounting
Robust MAP priors (<i>Schmidli et al., 2014</i>)	Summary counts	Multi-study $H \geq 3$	Bayes factor $m_k/m_V \implies \tilde{w}_R \rightarrow 1$	Cromwell's rule mixture with π_V
Pairwise shrinkage (<i>Röver & Friede, 2020</i>)	Summary contrast	Pairwise $H = 1$	Integrated variance τ $a_0 = \frac{\sigma_0^2}{\sigma_0^2 + 2\tau^2}$	Pre-adjustment required (symmetric offset)
Power likelihood + frugal (<i>Lin et al., 2026</i>)	Individual (IPD)	Pairwise or multi $H \geq 1$	Out-of-sample RCT $\text{ELPD}_{\text{cv}} \implies \hat{\eta} \in [0, 1]$	Frugal MSM copula + selective $\hat{\eta} \rightarrow 0$

The omnibus ELPD trade-off (Lin et al., 2026)

Because $\hat{\eta}$ optimizes joint predictive density over all variables rather than isolating $\tau(c)$, moderate confounding may fail to trigger predictive penalties, causing transient under-coverage before severe bias forces $\hat{\eta} \rightarrow 0$.

Summary of methodological evolution:

- Transition from static subjective power priors to dynamic, data-adaptive borrowing.
- Pairwise reference shrinkage resolves the single-historical-study ($H = 1$) bottleneck for contrast summaries.
- Combining power likelihoods with frugal parameterization unlocks granular, individual-level borrowing while protecting against observational misspecification.

Open research frontiers:

- *Contrast-targeted utility*: Formulating predictive scores focused strictly on the causal contrast $\tau(c)$ to eliminate transient under-coverage.
- *Semiparametric copulas*: Relaxing parametric copula assumptions.
- *Regulatory adoption*: Establishing formal guidelines for dynamic Bayesian borrowing in Phase III rare-disease and pediatric clinical trials.

Thank you!

Questions and discussion



Review paper & presentation material

github.com/ClaudiaFran/bayesian-data-fusion-review